INTERNAL · CALL PREP · ENCHANTED ROCK (EROCK) · OCT 26, 2026

BRIDGE POWER, THE FULL WIRE SCOPE

A modular natural-gas genset microgrid built the way ERock builds it: hundreds of ~450 kW gensets paralleled in blocks, stepped up and collected to the site switchgear. Every megawatt passes through 600 V leads first, so the cable story is mostly low-voltage, then MV collection.

GENSET
~450 kW natural gas, 480 V (assumed), ~680 A

→ GENERATOR LEADS 2 X 3 X 500 KCMIL

BLOCK SWITCHGEAR
480 V paralleling, 4 gensets per block (assumed), ~2,700 A

→ BLOCK SECONDARY 8 X 3 X 600 KCMIL

PAD-MOUNT STEP-UP
2,500 kVA, 480 V / 34.5 kV per block

→ MV COLLECTION LOOPS 1/C 500 KCMIL MV-105 35 KV

34.5 KV SWITCHGEAR
site collector, then the data centre or the utility point of interconnection

+ CONTROL AND SYNC PER GENSET · + INSTRUMENTATION · + FIBER RING TO THE MICROGRID CONTROLLER · + AUX POWER: JACKET-WATER HEATERS, CHARGERS · + GROUNDING GRID

CABLE FAMILY	WHERE IT RUNS	ASSUMED SIZE	SOUTHWIRE ANSWER	PER BLOCK (4 UNITS)	PER 100 MW	META-SCALE 366 MW
GENERATOR LEADS	genset to block switchgear	2 sets x 3 x 500 kcmil Cu, 600 V, + 1/0 ground	TC-ER or XHHW-2, SIMpull; paralleled sets on SIMpull reels	**3,200 ft**	**34 mi**	**124 mi**
BLOCK SECONDARY	switchgear to pad-mount	8 sets x 3 x 600 kcmil Cu + ground	TC-ER / XHHW-2 on SIMpull Stack reels (or bus duct)	**960 ft**	**10 mi**	**37 mi**
MV COLLECTION	pad-mounts to 34.5 kV switchgear	1/C 500 kcmil MV-105, 35 kV, 133 %	MV-105 EPR, pre-terminated with elbows	**450 ft**	**9 mi**	**32 mi**
AUX POWER	aux panel to each genset	3/C #10	TC-ER power	**400 ft**	**4 mi**	**16 mi**
CONTROL AND SYNC	genset to block controller	12/C #14	TC-ER control	**400 ft**	**4 mi**	**16 mi**
INSTRUMENTATION	gas, temperature, emissions	shielded pairs	PLTC / ITC	**400 ft**	**4 mi**	**16 mi**
FIBER	block ring to the microgrid controller	12-strand single-mode	fiber (confirm offer / partner)	**400 ft**	**4 mi**	**16 mi**
GROUNDING	block pad rings, equipment	4/0 bare Cu, #2 to each genset	bare copper	**420 ft**	**5 mi**	**16 mi**
Total			**copper, all families (Cu assumed)**	**~6,600 ft**	**~75 mi · ~390k lb Cu**	**~270 mi · ~1.4M lb Cu**

WHERE WE WIN

- **Standard blocks = standard kits.** Every block is the same: cut-to-length, tagged lead sets per genset, shipped per block.
- **Paralleled pulls.** ~1,600 lead pulls at 366 MW: SIMpull and paralleled reels cut the pull time.
- **Factory side.** 1 GW/yr of genset production in Texas: harness and enclosure wire in the factory (bucket 1).
- **Reel logistics** on a 30-acre site: staged delivery, reel return, job-site services.
- **MV pre-terminated** with elbows: faster energisation of each block.

ASK ON THE CALL

- Genset output voltage, kW per unit, gensets per block.
- Collection voltage: 13.8 or 34.5 kV? Duct bank, tray or direct burial?
- Who buys the cable: ERock, the EPC, or the utility owner (El Paso Electric for Meta)?
- Copper or aluminium on the leads and collection?
- Factory wiring inside the enclosure: who supplies it today?
- Schedule: Meta (366 MW, May 2027 target), Microsoft San Jose, the pipeline.

THEIR PUBLIC NUMBERS

813 GENSETS, META 366 MW

224 GENSETS, MICROSOFT 100 MW

~450 kW PER GENSET

1 GW/yr PRODUCTION CAPACITY

Estimate on an assumed design (480 V, 4 gensets per block, 2,500 kVA pad-mounts, 34.5 kV collection, 100 ft lead runs, ~1,500 ft MV home runs); not ERock data. Confirm on the call. Public sources: Microgrids Now (Meta, IPO), Microgrid Knowledge (San Jose), CPUC (448 kW units), mgrid.org (1 GW/yr).

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